

Foundations and Probability

CSC 5666: Advanced Machine Learning, Fall 2026
Professor Austin P. Wright: Course Notes

Abstract

Introduction to the fundamentals of Probability Theory and Bayesian Statistics the rest of the course will be built on. Closely follows Chapters 2–3 of Murphy’s textbook.

I Probability

In this course we are interested in learning from data, and the fundamental position from which learning is possible is through uncertainty. The best tool we have in understanding uncertainty is the theory of probability, which we will introduce in a semi-formal way here in order to give you at least the conceptual understanding of the mathematical objects we will be working with.

Probability spaces and random variables

Definition: Probability space

A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ consists of a sample space Ω of all possible outcomes of some experiment, an event space $\mathcal{F}$ of subsets of Ω closed under complements and countable unions, and a measure¹ $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ that assigns a size to each event.

The process of **sampling** from the probability space produces an outcome $o \in \Omega$, but we cannot say which specific outcome. Instead the probability measure is what determines which outcomes are more or less likely. All uncertainty in probability theory, and thus in all of our models, derives from this foundational uncertainty of outcomes of sampling based on a probability measure. This process is a sufficiently general and expressive source of uncertainty that we can then develop the theories of probability around different properties of objects that relate to or structure probability spaces.

An **event** $E \in \mathcal{F}$ is a set of outcomes, where if the outcome of a sample ($o \in \Omega$) is in E ($o \in E$), the event is considered to have occurred. A **random variable** $X : \Omega \rightarrow \mathcal{X}$ reads off a numerical or categorical summary of an outcome. We almost always work with the distribution X induces on $\mathcal{X}$ and never touch Ω directly, the same way nobody enumerates “every possible atmospheric state” when they check tomorrow’s forecast, instead we look at the distribution of likely temperatures.

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Textbook: §2.1, pp. 5–8

¹This course will not be going into the nuances of Measure Theory which are the “foundations of the foundations” for us since Measure theory is how Probability theory is built. However for the working computer scientist, we can stop the regress of foundations here.

We define the fundamental properties of a probability space using the Kolmogorov Axioms.²

Definition: Kolmogorov axioms

- $\mathbb{P}(E) \geq 0$.
- $\mathbb{P}(\Omega) = 1$.
- For disjoint $\{E_i\}$, $\mathbb{P}(\bigcup_i E_i) = \sum_i \mathbb{P}(E_i)$.

²Cox's theorem: any scheme for reasoning under uncertainty that satisfies a few common-sense desiderata is **forced** to reduce to the sum and product rules.

Discussion

Based on the Kolmogorov axioms, derive the values of:

1. $\mathbb{P}(\neg E) =$
2. $\mathbb{P}(A \cup B) =$

Discussion

Consider the case of sampling the values of two fair dice. One is green and the other is gold, and the result is always read in order of green then gold.

1. Define Ω as ordered outcomes. Would it be different with two green die?
2. List the event sets $S = \{\text{sum is 8}\}$ and $D = \{\text{at least one die is 3}\}$.
3. Compute $\mathbb{P}(S)$ and $\mathbb{P}(D)$, then locate $S \cap D$ before computing $\mathbb{P}(S \cup D)$.
4. Check the result two ways: inclusion–exclusion (i.e. event union) and direct counting. Which mistake does the second method expose?
5. Two dice give a small number of outcomes that is easy to list by hand. A poker hand is 5 cards from a 52-card deck: about 2.6 million possibilities. Nobody computes poker odds by listing hands one at a time. What do people (and computers) do instead?

Distributions and summaries

Once we have a random variable, we describe it directly on its range rather than returning to $\mathcal{F}$.

Textbook: §2.2, pp. 8–22

Definition: pmf, pdf, and cdf

For discrete X , the probability mass function is $p_{X(x)} = \mathbb{P}(X = x)$ and $\sum_x p_{X(x)} = 1$.

For continuous X , a probability density satisfies $p_{X(x)} \geq 0$ and $\int p_{X(x)} dx = 1$, with $\mathbb{P}(a < X \leq b) = \int_a^b p_{X(x)} dx$. **Note that a density value is not a probability.**

In either case, the cumulative distribution function is $F_{X(x)} = \mathbb{P}(X \leq x)$.

Definition: Expectation and variance

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x)p_{X(x)} & \text{discrete} \\ \int g(x)p_{X(x)} dx & \text{continuous} \end{cases} \quad (1)$$

We then define common statistics thusly:

- $\mu = \mathbb{E}[X]$
- $\mathbb{V}[X] = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
- $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$.

Using these definitions we can see some of the algebraic properties of expected values:

1. Linearity always holds for means, $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$
2. Variances add only when covariance vanishes: $\mathbb{V}[X + Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\text{Cov}(X, Y)$.³

³This is part of why we care so much about independence, it makes certain transformations possible.

Joint, marginal, and conditional distributions

A model of several variables starts from their **joint distribution** $p(x, y)$. We can then **marginalize** over a random variable to reconstruct individual distributions:

$$p(x) = \sum_y p(x, y) \quad \text{or} \quad p(x) = \int p(x, y) dy. \quad (2)$$

We then can define the **conditional probability** as the probability of some event occurring, *given that some other event is already known*. What this involves is considering the intersection of the events, and renormalizing to ensure the conditional probability meets the Kolmogorov axioms, that is conditioning forms a kind of “new probability space” where some of our uncertainty has been concretized.

Textbook: §2.1.5–2.1.6, pp. 7–8

Definition: Conditional probability

For $p(y) > 0$,

$$p(x|y) = \frac{p(x, y)}{p(y)}, \quad p(x, y) = p(x|y)p(y). \quad (3)$$

Conditioning selects outcomes compatible with y , then renormalizes.

Repeated use of the product rule gives the **chain rule**:

$$p(x_{1:N}) = p(x_1) \prod_{n=2}^N p(x_n | x_{1:n-1}). \quad (4)$$

The **law of total probability** follows by marginalization:

$$p(y) = \sum_x p(y|x)p(x). \quad (5)$$

Definition: Independence

X and Y are independent, $X \perp Y$, if and only if $p(x, y) = p(x)p(y)$ for every x, y .

They are conditionally independent given Z , $(X \perp Y) \mid Z$, if and only if $p(x, y|z) = p(x|z)p(y|z)$ for every relevant x, y, z .

Independence is a strong claim, in fact strictly speaking real variables are rarely exactly independent. Conditional independence is weaker, and it's what makes big joint distributions tractable because the factorization ensures that each distribution can be relatively small to keep track of, as the amount of data to represent a larger joint distribution increases combinatorially with the number of dependent variables.⁴

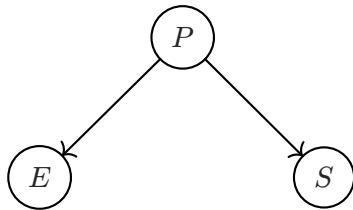


Figure 1: A common cause can induce dependence that disappears after conditioning.

⁴This is the reason we care about graphical models to express independence assumptions, as these assumptions are nearly always present and have a huge impact on how we can model distributions.

Discussion

Let P be preparation, E exam score, and S sleep the night before.

1. Write the unrestricted factorization $p(p, e, s)$ using the chain rule.
2. Read the diagram above as representing the joint as

$$p(p, e, s) = p(p)p(e|p)p(s|p) \quad (6)$$

Which term was simplified, and what assumption licenses dropping it?

3. Decide whether $E \perp S$ and whether $E \perp S \mid P$ must hold.
4. Give one omitted variable that would make the conditional-independence claim implausible (something that affects both sleep and exam performance directly). Redraw the model.

Bayes' rule

Based on our definitions so far developed, we can do some simple algebra to derive some interesting relationships within the joint $p(x, y)$.

Textbook: §2.1.6, p. 8

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} = \frac{p(y|x)p(x)}{\sum_{x'} p(y|x')p(x')}. \quad (7)$$

Despite this simplicity, this equation forms the fundamental structure of Bayesian modeling, and is sometimes called “the most important equation in all of artificial intelligence”.

Definition: Bayesian vocabulary

$$\underbrace{p(x|y)}_{\text{posterior}} = \frac{\underbrace{p(y|x)}_{\text{likelihood}} \underbrace{p(x)}_{\text{prior}}}{\underbrace{p(y)}_{\text{evidence}}}. \quad (8)$$

As a function of x , $p(y|x)$ need not normalize. Therefore $p(x|y) \propto p(y|x)p(x)$ is often the useful form.

Discussion

A disease has prevalence 0.01; a test has sensitivity 0.95 and false-positive rate 0.05.

1. Predict whether $\mathbb{P}(\text{disease}|\text{positive})$ is above or below $\frac{1}{2}$.
2. Recompute using Bayes' rule and reconcile the two answers.
3. Which single change, to prevalence, sensitivity, or specificity, would most improve the posterior here?

Useful distribution families

Picking a likelihood means picking a family for each observed or latent quantity. Start from **support** which is what values are even possible, then worry about shape.

Textbook: §2.2, pp. 8–22

- Bernoulli(θ): one binary outcome, Binomial(N, θ): multiple trials count.
- Categorical($\vec{\theta}$): one of K classes; Multinomial($N, \vec{\theta}$): class counts.
- Poisson(λ): count in a fixed rate, mean equal to variance
- Exponential(λ): positive waiting time between events.
- Beta(α, β): probability in $[0, 1]$; Dirichlet($\vec{\alpha}$): probability vector.
- Gaussian $\mathcal{N}(\mu, \sigma^2)$
- Multivariate Gaussian $\mathcal{N}(\vec{\mu}, \Sigma)$

Discussion

A city's 911 dispatch center wants a likelihood for:

1. whether a given call is a false alarm (yes/no);
2. calls received per hour;
3. time until the next call arrives;
4. call counts split across four neighborhoods.

For each, name its parameter and one implied assumption. Then diagnose: which of these breaks under a holiday surge, a lull between shifts, or a single caller placing repeated calls, and what would you switch to?

Gaussian systems

Textbook: §2.3, pp. 22–33

The Gaussian is mathematically convenient enough that whole model families are built from nothing else.

Definition: Multivariate Gaussian

For $\vec{x} \in \mathbb{R}^D$,

$$\mathcal{N}(\vec{x}|\vec{\mu}, \Sigma) = (2\pi)^{-\frac{D}{2}} |\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\vec{x} - \vec{\mu})^T \Sigma^{-1} (\vec{x} - \vec{\mu})\right). \quad (9)$$

Σ is symmetric positive semidefinite; its diagonal stores variances and its off-diagonal stores covariances.

- If $\vec{x} \sim \mathcal{N}(\vec{\mu}, \Sigma)$ and $\mathbf{y} = A\vec{x} + \mathbf{b} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, Q)$ independent, then $\mathbf{y} \sim \mathcal{N}(A\vec{\mu} + \mathbf{b}, A\Sigma A^T + Q)$.
- Marginals and conditionals of a joint Gaussian are Gaussian. Conditioning updates both mean and covariance analytically.⁵

Zero covariance implies independence for jointly Gaussian variables, but this does not hold in general.

⁵This pair of facts, run over a continuum of indices instead of a finite vector, is the entire Gaussian-process regression algorithm — later this course.

II Statistics: Learning from Data

While probability distributions allow us to talk about, given a model, what data is likely to appear. Statistics reverses it, given data $\mathcal{D} = \{y_1, \dots, y_N\}$, what parameters produced it?

Textbook: Chapter 3, pp. 63–142

Bayesian inference

Textbook: §3.2, pp. 63–72

Definition: Bayesian inference

$$p(\theta|\mathcal{D}) = \frac{p(\theta)p(\mathcal{D}|\theta)}{p(\mathcal{D})} = \frac{p(\theta)p(\mathcal{D}|\theta)}{\int p(\theta')p(\mathcal{D}|\theta')d\theta'}. \quad (10)$$

Note that the posterior is a **distribution** and not a single estimate. From the distribution we can estimate:

- a point summary: posterior mean, median, or MAP;
- uncertainty: credible intervals and joint posterior geometry;
- prediction: average over parameter uncertainty;
- evidence: compare complete probabilistic models.

Four different questions, one object: best guess, how sure, what happens next, is this model any good.

Running example: tossing a coin

We work this by hand because every quantity above has a closed form here.

Let $y_n \in \{0, 1\}$, $\theta = \mathbb{P}(y_n = 1|\theta)$, and let $N_1 = \sum_n y_n$, $N_0 = N - N_1$. The likelihood is

$$p(\mathcal{D}|\theta) = \theta^{N_1} (1 - \theta)^{N_0}. \quad (11)$$

With prior $\theta \sim \text{Beta}(\alpha_0, \beta_0)$,

$$p(\theta|\mathcal{D}) = \text{Beta}(\hat{\alpha}, \hat{\beta}), \quad \hat{\alpha} = \alpha_0 + N_1, \quad \hat{\beta} = \beta_0 + N_0. \quad (12)$$

The update adds observed counts to prior pseudo-counts.

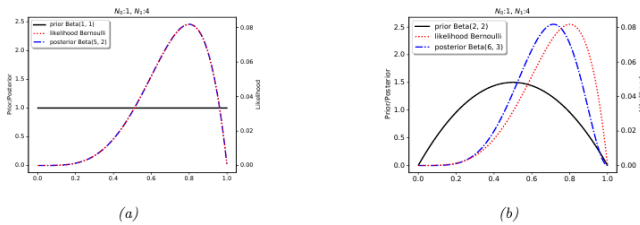


Figure 2: Uniform and weakly informative beta priors updated by the same Bernoulli data. Reproduced from Murphy, Textbook Figure 3.1, p. 65 (CC BY-NC-ND).

$$\mathbb{E}[\theta|\mathcal{D}] = \frac{\hat{\alpha}}{\hat{\alpha} + \hat{\beta}}, \quad (13)$$

$$\text{MAP}(\theta) = \frac{\hat{\alpha} - 1}{\hat{\alpha} + \hat{\beta} - 2} \quad (14)$$

when $\hat{\alpha}, \hat{\beta} > 1$, and

$$\mathbb{V}[\theta|\mathcal{D}] = \frac{\hat{\alpha}\hat{\beta}}{(\hat{\alpha} + \hat{\beta})^2(\hat{\alpha} + \hat{\beta} + 1)}. \quad (15)$$

Definition: Credible interval

A $(1 - \delta)$ credible set C satisfies $\mathbb{P}(\theta \in C|\mathcal{D}) = 1 - \delta$. This is a probability statement about θ conditional on the observed data.

Prediction and evidence

When discussing uncertainty, we must distinguish between what the source of our uncertainty is. We can have both **aleatoric** uncertainty (rain is genuinely random) and **epistemic** uncertainty (you're unsure your forecast model is any good). More data shrinks the second but not the first. Bayesian approaches that utilize good prior distributions allow us to include epistemic uncertainty in our analysis.

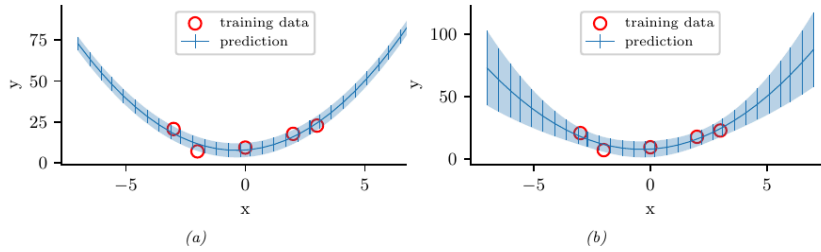


Figure 3: Aleatoric uncertainty stays narrow away from the data; Bayesian prediction widens as parameter uncertainty grows. Reproduced from Murphy, Textbook

Figure 3.3, p. 71 (CC BY-NC-ND).

Exchangeability

An interesting philosophical question is: where do priors come from, given that they refer to parameters which are just abstract quantities in a model, and not directly observable. A fundamental result, known as de Finetti's theorem, explains how they are related to our beliefs about observable outcomes.

Definition: Exchangeability

$X_1, \dots, X_N$ are infinitely exchangeable if their joint distribution is unchanged by permutation.

Definition: de-Finetti's Theorem

$X_1, \dots, X_N$ are infinitely exchangeable if and only if we have:

$$p(x_{1:N}) = \int \prod_{n=1}^N p(x_n|\theta)p(\theta) d\theta. \quad (16)$$

Where θ is some hidden, common, random variable.

We often interpret θ as a parameter. The theorem tells us that, if our data is exchangeable, then there must exist a parameter θ , and a likelihood $p(x_i|\theta)$, and a prior $p(\theta)$. Thus the Bayesian approach follows automatically from exchangeability.

Conjugate Priors

Textbook: §3.4, pp. 83–101

Definition: Conjugate prior

A prior family is conjugate to a likelihood when the posterior remains in the same family. That is if $p(\theta) \in \mathcal{F}$ then $p(\theta | \mathcal{D}) \in \mathcal{F}$. Thus $\mathcal{F}$ is closed under Bayesian updating, and so inference reduces to updating hyperparameters.

Most relevant for us is that the multivariate normal distribution with known covariance is *conjugate to itself*. You can derive conjugacy for common distributions or [look them up](#). While conjugate priors can help some computations, they are not an excuse to make poorly justified modeling choices.

Selecting and checking priors

Choosing a prior is a central part of the model. Specifying exactly how much uncertainty you have to start is a surprisingly difficult task, and saying: “I don’t know, so I’ll be vague” is itself a claim with consequences, and should be defended like any other modeling decision.

Textbook: §§3.2.3, 3.4.6–3.5, pp. 71, 99–107

Some common language to describe kinds of priors include:

- **informative:** encodes substantial external evidence⁶
- **weakly informative:** rules out implausible extremes while leaving many values possible
- **regularizing:** stabilizes estimation toward a simpler region
- **conjugate:** chosen partly for algebraic convenience
- **hierarchical/empirical:** learns shared prior structure from related data

⁶The opposite of this, noninformative, is not literal since every prior privileges some scale or parameterization. Prefer default, diffuse, or minimally informative, and state the criterion used.

Maximum entropy priors

A natural way to define an uninformative prior is to use one that has maximum entropy, since it makes the least commitments to any particular value in the state space.⁷

Definition: Maximum entropy prior

Among distributions satisfying some stated constraints $\mathbb{E}[f_k(\theta)] = F_k$, choose

$$p^* = \arg \max_p \mathbb{H}(p). \quad (17)$$

Max entropy allows us to formalize “what’s the least extra information beyond exactly these constraints.” Useful examples are that a mean-and-variance constraint gives a Gaussian; a mean-only constraint on $[0, \infty)$ gives an Exponential.

⁷ We will discuss entropy in more detail later on.

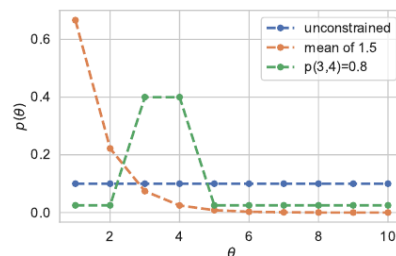


Figure 4: Maximum entropy is relative to constraints: none, a fixed mean, or concentrated mass on selected values. Reproduced from Murphy, Textbook Figure 3.12, p. 104 (adapted there from MKL11; CC BY-NC-ND).

Maximum entropy makes no commitments **beyond the declared constraints**. It does not decide which support, coordinates, or constraints are appropriate.

*Jeffreys priors***Definition: Jeffreys prior**

$$p_{J(\theta)} \propto \sqrt{\det \mathcal{I}(\theta)}, \quad (18)$$

where $\mathcal{I}(\theta)$ is Fisher information. The prior is invariant under smooth one-to-one reparameterization.

Consider the 1d case:

$$\begin{aligned} p_{\varphi(\varphi)} &= p_{\theta(\theta)} \left| \frac{d\theta}{d\varphi} \right| \\ &\propto \sqrt{F(\theta) \left(\frac{d\theta}{d\varphi} \right)^2} = \sqrt{\mathbb{E} \left[\left(\frac{d \log p(x|\theta)}{d\theta} \right)^2 \right] \left(\frac{d\theta}{d\varphi} \right)^2} \\ &= \sqrt{\mathbb{E} \left[\left(\frac{d \log p(x|\theta)}{d\theta} \frac{d\theta}{d\varphi} \right)^2 \right]} = \sqrt{\mathbb{E} \left[\left(\frac{d \log p(x|\varphi)}{d\varphi} \right)^2 \right]} \\ &= \sqrt{F(\varphi)} \end{aligned} \quad (19)$$

Note that there is no parameterization-free notion of “knowing nothing.” In more than one dimension, Jeffreys priors get unwieldy and are rarely used directly; people fall back on priors motivated by scale instead.

Known symmetries give related defaults:

- translation invariance for location: $p(\mu) \propto 1$;
- scale invariance: $p(\sigma) \propto \frac{1}{\sigma}$;
- a reference prior maximizes expected information gain $\mathbb{E}_{\mathcal{D}} [D_{\text{KL}}(p(\theta|\mathcal{D}) \parallel p(\theta))]$ and equals Jeffreys in one dimension.